

Given: Due:	Further Maths (Core Pure) HW 25	MARK SCHEME
Chapter 9: Vector lines, planes & distance		
Q1. The line l has equation $\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$ The plane Π has equation $\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -7$ Determine whether the line l intersects Π at a single point, or lies in Π, or is parallel to Π without intersecting it. (Total for question = 5 marks)		

This question tested a geometrical understanding of the interaction of planes and lines, and was unfamiliar to many candidates. However, most candidates managed at least 3 marks on this question, with a third or so achieving full marks.

Despite there being a variety of methods, the most common one was via the main scheme approach, and this was also the most successfully attempted. Mixed approaches (the main scheme and Alt 1) tended to revert back to the main scheme at some point. The second alternative was very rare, and commonly unsuccessfully completed.

The majority of candidates did achieve the equation of the line in parametric form, though a few managed to mix up the format of the equation. For some, this was as far as they proceeded, not knowing what to do with it. But most candidates did proceed to perform substitute into the plane equation. Mistakes with signs were quite common at this stage, particularly the loss of the $-$ sign from the -7 , but these often still led to a contradiction.

When a correct contradiction was achieved, candidates generally did not realise that this eliminated two of the three given possibilities and so went on to try and prove the line and plane were parallel using the scalar product of directions. Likewise, many started on this route before reverting to the main scheme when needing to determine if they intersected, rather than simply testing one point once they knew line and plane were parallel. The geometry of the situation was not well understood, but the carrying out of procedures was done well.

Question	Scheme	Marks	AOs
	$(\mathbf{r} \Rightarrow \begin{pmatrix} -2+\lambda \\ 5-\lambda \\ 4-3\lambda \end{pmatrix} \text{ or } \begin{pmatrix} -2 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} \text{ (oe)})$	M1	1.1b
	So meet if $\begin{pmatrix} -2+\lambda \\ 5-\lambda \\ 4-3\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = -7 \Rightarrow (-2+\lambda) \times 1 + (5-\lambda) \times -2 + (4-3\lambda) \times 1 = -7$	M1 A1	3.1a 1.1b
	$\Rightarrow 0\lambda - 8 = -7 \Rightarrow -8 = -7$ a contradiction so no intersection	A1ft	2.3
	Hence l is parallel to Π but not in it.	A1cso	3.2a
		(5)	
	(5 marks)		



Notes

	M1 Forms a parametric form for the line. Allow one slip. M1 Substitutes into the equation of the plane to an equation in λ. May use Cartesian form of plane to substitute into. A1 Correct equation in λ A1ft Simplifies and derives a contradiction and deduces line and plane do not meet. Follow through in their initial equation in λ so - contradiction so no intersection if λ disappears and constants unequal - line lies in plane if a tautology is arrived at - meet in a point if a solution for λ is found. But do not allow for incorrect simplification from a correct initial equation in λ Note that a miscopy/misread of 7 instead of -7 can therefore score a maximum of M1M1A0A1A0. A1cso Correct deduction from correct working. This may be seen two separate statements in their working. You may see attempts at showing the line is parallel before/after deducing there is no intersection.		
Alt 1	Note that some may attempt a mix of the main scheme and Alt 1. Mark under main scheme unless Alt 1 would score higher.		
	$\begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 1 \times 1 + (-1) \times (-2) + (-3) \times 1 = 0$	M1	3.1a
	Hence l is parallel to Π	A1	1.1b
	$(-2, 5, 4)$ on l , but $(1)(-2) + (-2)(5) + 1(4) = -8$	M1	1.1b
	$-8 \neq -7$ so $(-2, 5, 4)$ is not on the plane.	A1ft	2.3
	Hence l is (parallel to Π but) not in the plane.	A1cso	3.2a
		(5)	
(5 marks)			

Alt 1 Notes

M1	Attempts the dot product between the two direction vectors.
A1	Shows dot product is zero and makes the correct deduction that line is parallel to plane.
M1	Finds a point on l and substitutes into the equation of Π (vector or Cartesian)
A1ft	Simplifies and derives a contradiction – follow through their equation, so if arrive at a tautology, they should deduce the line is in the plane.
A1cso	Correct deduction from correct working but may be split across working.

Question	Scheme	Marks	AOs
Alt 2	Attempts to solve $\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$ and $x-2y+z=-7$ simultaneously – eliminates one variable for M mark.	M1	3.1a
	e.g. $y = -(x+2)+5 = -x+3 \Rightarrow x-2(-x+3)+z=-7 \Rightarrow 3x+z=-1$ (oe)	A1	1.1b
	Solves reduced equations, e.g. $-3(x+2) = z-4 \Rightarrow 3x+z=-2$ and $3x+z=-1 \Rightarrow (3x+z)-(3x+z) = -2-(-1)$	M1	1.1b
	$\Rightarrow 0 = -1$ a contradiction so no intersection	A1ft	2.3
	Hence l is parallel to Π but not in it.	A1cso	3.2a
		(5)	
(5 marks)			

Alt 2 notes

M1	Attempts to solve the Cartesian equation of the line and plane, using the plane equation to eliminate one variable for the M.
A1	Correct elimination of their chosen variable. (E.g may see $3-3y+z=-7$ or $-2x-2y-2=-7$ etc)
M1	Solves the reduced equations in two variables...
A1ft	... and derives a contradiction/line and plane do not meet. Follow through their result, so may reach a tautology and deduce lies in plane, or find single solution and deduce meet in a point.
A1cso	Correct deduction from correct working.

Q2. The line l passes through the point $P(2,1,3)$ and is perpendicular to the plane Π whose vector equation is $\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = 3$

Find

(a) a vector equation of the line l ,

(2)

(b) the position vector of the point where l meets Π .

(c) Hence find the perpendicular distance of P from Π .

In question (a) the vast majority of the students correctly identified the direction vector of the line and thus confidently formed the vector equation of the line, usually in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$. In question (b), the substitution of the line, once expressed in parametric form, into the given equation of the plane and a solution determined for λ , giving the required point, was well known by many of the students and most scored full marks in this part. However, in question (c), a significant minority did not use this point to determine the perpendicular distance of P from the plane as requested, with many resorting to using the result given in the formula book, which scored no marks.

Question	Scheme		Marks
	$P(2, 1, 3)$ and $\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = 3$		
(a)	l is parallel to $\mathbf{i} - 2\mathbf{j} - \mathbf{k}$	An appreciation that $\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ is the direction of the line (may be implied).	M1
	$\mathbf{r} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k} + t(\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ or $(\mathbf{r} - (2\mathbf{i} + \mathbf{j} + 3\mathbf{k})) \times (\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = \mathbf{0}$ or $\mathbf{r} \times (\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = (2\mathbf{i} + \mathbf{j} + 3\mathbf{k}) \times (\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ or $\mathbf{r} \times (\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = 5\mathbf{i} + 5\mathbf{j} - 5\mathbf{k}$	A correct vector <u>equation</u> in any form. (Allow any multiple of the direction vector.)	A1
			(2)
(b)	$\begin{pmatrix} 2+t \\ 1-2t \\ 3-t \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = 3$	Substitutes a parametric form of their line from part (a) into the equation of the plane. This statement is sufficient.	M1
	$2+t-2(1-2t)-(3-t)=3$	Correct equation (allow unsimplified)	A1
	$2+t-2+4t-3+t=3 \Rightarrow t = \dots\dots$	Solves to find a value for t Dependent on the first M	dM1
	$t=1 \Rightarrow l$ meets Π at $\mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$	Correct position vector (allow as coordinates (3, -1, 2))	A1
			(4)
(c) Way 1	$PQ = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} - (2\mathbf{i} + \mathbf{j} + 3\mathbf{k}) $		
	$= \mathbf{i} - 2\mathbf{j} - \mathbf{k} = \sqrt{1^2 + 2^2 + 1^2}$	Attempts the vector PQ or QP and correct Pythagoras.	M1
	$= \sqrt{6}$	Allow awrt 2.45 or $\frac{6}{\sqrt{6}}$	A1
			(2)
(c) Way 2	$t = "1" \Rightarrow \overrightarrow{PQ} = "1" \times (\mathbf{i} - 2\mathbf{j} - \mathbf{k})$		
	$= \mathbf{i} - 2\mathbf{j} - \mathbf{k} = \sqrt{1^2 + 2^2 + 1^2}$	Attempts the vector PQ using their value for t and their normal vector and correct Pythagoras.	M1
	$= \sqrt{6}$	Allow awrt 2.45 or $\frac{6}{\sqrt{6}}$	A1
			(2)
			Total 8

Q3. With respect to a fixed origin O , the line l has equation

$$\mathbf{r} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \text{ where } \lambda \text{ is a scalar parameter.}$$

The point A lies on l and has coordinates $(3, -2, 6)$.

The point P has position vector $(-pi + 2pk)$ relative to O , where p is a constant.

Given that vector $\overrightarrow{PA}$ is perpendicular to l ,

(a) find the value of p .



Given also that B is a point on l such that $\angle BPA = 45^\circ$;

(b) find the coordinates of the two possible positions of B .

(5)

In general, this was the most poorly answered question on the paper with about 25% of candidates failing to score. Some candidates did not seem to have a firm grasp of what was required and many produced pages of irrelevant working. This question did discriminate well between candidates of average to higher abilities, with about 54% of candidates gaining at least 4 of the 9 marks available and only about 10% of candidates gaining all 9 marks. Part (a) was found to be fairly accessible, and part (b) was challenging to all but the most able candidates. Many were unable to think about the question logically or produce a clear diagram and establish the relationship between the length of AB (and/or PB) and the length of PA .

$$\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

In part (a), the vector $\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ (or $\overrightarrow{OA}$) was usually found, although sign slips, adding to $\overrightarrow{OA}$ and mixing up of the i , j and k components of were common errors. Many candidates found the correct value of p by

$$\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

applying the correct scalar product between $\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ (or $\overrightarrow{OA}$) and the direction vector $2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and setting the result equal to 0, although some candidates used $\overrightarrow{OA} \cdot 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ or $13\mathbf{i} + 8\mathbf{j} + \mathbf{k}$ instead of $2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$. Other errors included taking the dot product between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ deducing $p = -1$ from a correct $4p - 4 = 0$.

Those candidates who attempted part (b) usually managed to find the magnitude of PA and many drew a diagram of triangle PAB correctly and deduced $PA = AB$. From this point, however, many candidates did not know how to proceed further, resulting in a lot of incorrect work which yielded no further marks. Some candidates, however, were able to form a correct equation in order to find both values of λ . It was unfortunate that a few, having found the correct values of $\lambda = -3$ and $\lambda = -7$ then substituted these into instead of the equation for the line l . The most popular method for finding correct values of λ was for candidates to form and solve a Pythagorean equation in λ of $AB = 6$ or $AB^2 = 36$. Other successful methods for finding λ

$$\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

included solving $PB = 6\sqrt{2}$ or solving a dot product equation between either $\begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ and $\overrightarrow{AB}$ or $\overrightarrow{AB}$ and $\overrightarrow{AB}$.

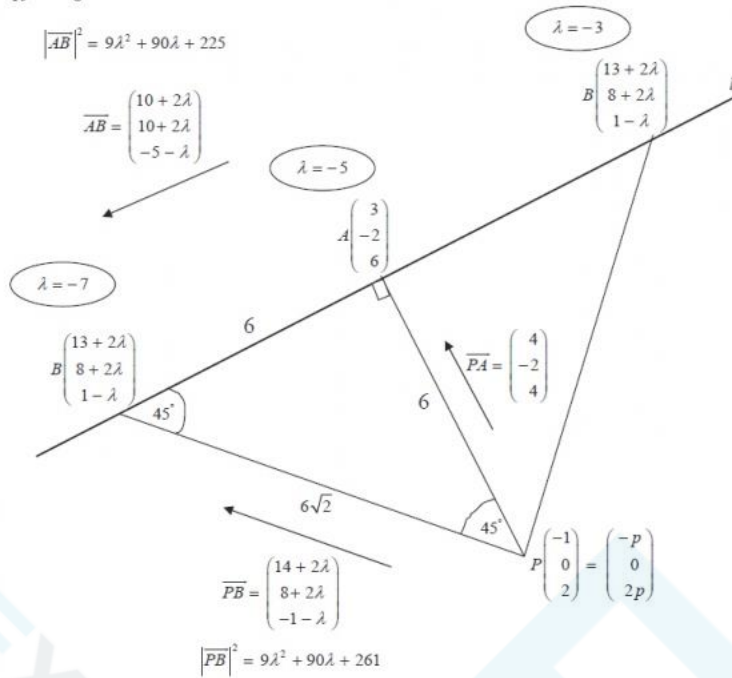
Few candidates realised that the length AB was twice the length of the direction vector of the line l and applied twice the direction vector $2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ in either direction from A in order to find both positions for B .



Question Number	Scheme	Marks
(a)	$l: \mathbf{r} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \quad A(3, -2, 6), \quad \overrightarrow{OP} = \begin{pmatrix} -p \\ 0 \\ 2p \end{pmatrix}$ $\{\overrightarrow{PA}\} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} - \begin{pmatrix} -p \\ 0 \\ 2p \end{pmatrix} \quad \left \quad \{\overrightarrow{AP}\} = \begin{pmatrix} -p \\ 0 \\ 2p \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} \right.$ $= \begin{pmatrix} 3+p \\ -2 \\ 6-2p \end{pmatrix} \quad \left \quad = \begin{pmatrix} -3-p \\ 2 \\ 2p-6 \end{pmatrix} \right.$ $\begin{pmatrix} 3+p \\ -2 \\ 6-2p \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = 6 + 2p - 4 - 6 + 2p = 0$ $p = 1$	Finds the difference between $\overrightarrow{OA}$ and $\overrightarrow{OP}$. Ignore labelling. M1 Correct difference. A1 See notes. M1
(b)	$ \overrightarrow{AP} = \sqrt{4^2 + (-2)^2 + 4^2} \quad \text{or} \quad \overrightarrow{AP} = \sqrt{(-4)^2 + 2^2 + (-4)^2}$ So, PA or $AP = \sqrt{36}$ or 6 cao It follows that, $AB = "6" \{ = PA \}$ or $PB = "6\sqrt{2}" \{ = \sqrt{2} PA \}$ {Note that $AB = "6" = 2$ (the modulus of the direction vector of l)} $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} \pm 2 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \quad \text{or}$ $\overrightarrow{OB} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} - 7 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 7 \\ 2 \\ 4 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ -6 \\ 8 \end{pmatrix}$	See notes. M1 Uses a correct method in order to find both possible sets of coordinates of B. M1 Both coordinates are correct. A1 cao
[4]		
[5]		
9		
Notes for Question		
(a)	M1: Finds the difference between $\overrightarrow{OA}$ and $\overrightarrow{OP}$. Ignore labelling. If no "subtraction" seen, you can award M1 for 2 out of 3 correct components of the difference. A1: Accept any of $\begin{pmatrix} 3+p \\ -2 \\ 6-2p \end{pmatrix}$ or $(3+p)\mathbf{i} - 2\mathbf{j} + (6-2p)\mathbf{k}$ or $\begin{pmatrix} -3-p \\ 2 \\ 2p-6 \end{pmatrix}$ or $(-3-p)\mathbf{i} + 2\mathbf{j} + (2p-6)\mathbf{k}$	

Notes for Question Continued	
(a)	M1: Applies the formula $\overrightarrow{PA} \cdot \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ or $\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ correctly to give a linear equation in p which is set equal to zero. Note: The dot product can also be with $\pm k \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$. Eg: Some candidates may find $\begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} = \begin{pmatrix} 10 \\ 10 \\ -5 \end{pmatrix}, \text{ for instance, and use this in their dot product which is fine for M1.}$ A1: Finds $p = 1$ from a correct solution only. Note: The direction of subtraction is not important in part (a).
(b)	M1: Uses their value of p and Pythagoras to obtain a numerical expression for either AP or PA or AP^2 or PA^2. Eg: PA or $AP = \sqrt{4^2 + (-2)^2 + 4^2}$ or $\sqrt{(-4)^2 + 2^2 + (-4)^2}$ or $\sqrt{4^2 + 2^2 + 4^2}$ or PA^2 or $AP^2 = 4^2 + (-2)^2 + 4^2$ or $(-4)^2 + 2^2 + (-4)^2$ or $4^2 + 2^2 + 4^2$ A1: AP or $PA = \sqrt{36}$ or 6 cao or $AP^2 = 36$ cao B1ft: States or it is clear from their working that $AB = "6" \{ = \text{their evaluated } PA \}$ or $PB = "6\sqrt{2}" \{ = \sqrt{2} (\text{their evaluated } PA) \}$. Note: So a correct follow length is required here for either AB or PB using their evaluated PA. Note: This mark may be found on a diagram. Note: If a candidate states that $\overrightarrow{AP} = \overrightarrow{AB}$ and then goes on to find $\overrightarrow{AP} = 6$ then the B1 mark can be implied. IMPORTANT: This mark may be implied as part of expressions such as: $\{AB = \sqrt{(10+2\lambda)^2 + (10+2\lambda)^2 + (-5-\lambda)^2} = 6 \text{ or } \{AB^2 = (10+2\lambda)^2 + (10+2\lambda)^2 + (-5-\lambda)^2 = 36$ $\text{or } \{PB = \sqrt{(14+2\lambda)^2 + (8+2\lambda)^2 + (-1-\lambda)^2} = 6\sqrt{2} \text{ or } \{PB^2 = (14+2\lambda)^2 + (8+2\lambda)^2 + (-1-\lambda)^2 = 72$ M1: Uses a full method in order to find both possible sets of coordinates of B. Eg 1: $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} \pm 2 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ Eg 2: $\overrightarrow{OB} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} - 7 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ Note: If a candidate achieves at least one of the correct $(7, 2, 4)$ or $(-1, -6, 8)$ then award SC M1 here. Note: $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} - 7 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ is M0. A1: For both $(7, 2, 4)$ and $(-1, -6, 8)$. Accept vector notation or $\mathbf{i}, \mathbf{j}, \mathbf{k}$ notation. Note: All the marks are accessible in part (b) if $p = 1$ is found from incorrect working in part (a). Note: Imply M1A1B1 and award M1 for candidates who write: $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} \pm 2 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$, with little or no earlier working.

Helpful Diagram!



- (b) **Way 2:** Setting $AB = "6"$ or $AB^2 = "36"$ **Note:** It is possible for you to apply the main scheme for Way 2.
 $\{AB = "6" \Rightarrow AB^2 = "36" \Rightarrow (10+2\lambda)^2 + (10+2\lambda)^2 + (-5-\lambda)^2 = "36"$ **B1ft** could be implied here.
 $9\lambda^2 + 90\lambda + 225 = 36 \Rightarrow 9\lambda^2 + 90\lambda + 189 = 0$
 $\lambda^2 + 10\lambda + 21 = 0 \Rightarrow (\lambda+3)(\lambda+7) = 0$
 $\lambda = -3, -7$
 Then apply final M1 A1 as in the original scheme. | ... M1 A1
- (b) **Way 3:** Setting $PB = "6\sqrt{2}"$ or $PB^2 = "72"$ **Note:** It is possible for you to apply the main scheme for Way 3.
 $\{PB = "6\sqrt{2}" \Rightarrow PB^2 = "72" \Rightarrow (14+2\lambda)^2 + (8+2\lambda)^2 + (-1-\lambda)^2 = "72"$ **B1ft** could be implied here.
 $9\lambda^2 + 90\lambda + 261 = 72 \Rightarrow 9\lambda^2 + 90\lambda + 189 = 0$
 $\lambda^2 + 10\lambda + 21 = 0 \Rightarrow (\lambda+3)(\lambda+7) = 0$
 $\lambda = -3, -7$
 Then apply final M1 A1 as in the original scheme. | ... M1 A1



Notes for Question Continued	
(b)	(You need to be convinced that a candidate is applying this method before you apply the Mark Scheme for Way 4). Way 4: Using the dot product formula between $\overrightarrow{PA}$ and $\overrightarrow{PB}$, ie: $\cos 45^\circ = \frac{\overrightarrow{PA} \cdot \overrightarrow{PB}}{ \overrightarrow{PA} \overrightarrow{PB} }$. $\overrightarrow{PA} \cdot \overrightarrow{PB} = \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 14+2\lambda \\ 8+2\lambda \\ -1-\lambda \end{pmatrix} = 56+8\lambda-16-4\lambda-4-4\lambda = 36$ $\{\cos 45^\circ\} = \frac{1}{\sqrt{2}} = \frac{36}{6\sqrt{9\lambda^2+90\lambda+261}}$ $\frac{1}{2} = \frac{36}{9\lambda^2+90\lambda+261}$ $9\lambda^2+90\lambda+261=72 \Rightarrow 9\lambda^2+90\lambda+189=0$ $\lambda^2+10\lambda+21=0 \Rightarrow (\lambda+3)(\lambda+7)=0$ $\lambda = -3, -7$ Then apply final M1 A1 as in the original scheme. ... M1 A1
(b)	(You need to be convinced that a candidate is applying this method before you apply the Mark Scheme for Way 5). Way 5: Using the dot product formula between $\overrightarrow{AB}$ and $\overrightarrow{PB}$, ie: $\cos 45^\circ = \frac{\overrightarrow{AB} \cdot \overrightarrow{PB}}{ \overrightarrow{AB} \overrightarrow{PB} }$. Attempts the dot product formula between $\overrightarrow{AB}$ and $\overrightarrow{PB}$. M1 Correct statement with $\overrightarrow{AB}$ and $\overrightarrow{PB}$ simplified as shown. A1 Either $\overrightarrow{AB} = \sqrt{9\lambda^2+90\lambda+225}$ or $\overrightarrow{PB} = \sqrt{9\lambda^2+90\lambda+261}$ B1 $\{\cos 45^\circ\} = \frac{1}{\sqrt{2}} = \frac{140+20\lambda+28\lambda+4\lambda^2+80+20\lambda+16\lambda+4\lambda^2+5+5\lambda+\lambda+\lambda^2}{\sqrt{9\lambda^2+90\lambda+225} \sqrt{9\lambda^2+90\lambda+261}}$ $\{\cos 45^\circ\} = \frac{1}{\sqrt{2}} = \frac{9\lambda^2+90\lambda+225}{\sqrt{9\lambda^2+90\lambda+225} \sqrt{9\lambda^2+90\lambda+261}}$ $\frac{1}{2} = \frac{(9\lambda^2+90\lambda+225)^2}{(9\lambda^2+90\lambda+225)(9\lambda^2+90\lambda+261)}$ $\frac{1}{2} = \frac{(9\lambda^2+90\lambda+225)}{(9\lambda^2+90\lambda+261)}$ $9\lambda^2+90\lambda+261 = 2(9\lambda^2+90\lambda+225) \Rightarrow 9\lambda^2+90\lambda+189=0$ $\lambda^2+10\lambda+21=0 \Rightarrow (\lambda+3)(\lambda+7)=0$ $\lambda = -3, -7$ Then apply final M1 A1 as in the original scheme. ... M1 A1

Notes for Question Continued	
(b)	Way 6: $\overrightarrow{PA} = \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \text{ and direction vector of } l \text{ is } \mathbf{d} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ So, $\overrightarrow{PA} = 2 \mathbf{d}$ or $PA = 2 \mathbf{d}$ M1 A1 B1 A correct statement relating these distances (and not vectors) M1 A1 B1 Apply final M1 A1 as in the original scheme. ... M1 A1 Note: $\overrightarrow{PA} = 2\mathbf{d}$ with no other creditable working is M0A0B0... Note: $\overrightarrow{PA} = 2\mathbf{d}$, followed by $\overrightarrow{OB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} \pm 2 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ is M1A1B1M1 and the final A1 mark is for both sets of correct coordinates.

Q4. The plane Π_1 has vector equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = 5$

The plane Π_2 has vector equation $\mathbf{r} \cdot \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} = 7$

(a) Find a vector equation for the line of intersection of Π_1 and Π_2 , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$ where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors and λ is a scalar parameter.

(6)

The plane Π_3 has cartesian equation

$$x - y + 2z = 31$$

(b) Using your answer to part (a), or otherwise, find the coordinates of the point of intersection of the planes Π_1 , Π_2 and Π_3

(3)

In question (a), students used a variety of methods to find an equation for the line of intersection of the two planes. Perhaps the most common approach was to attempt the Cartesian equation of the line by expressing one of x or y or z in terms of the other variables from the Cartesian equations of the two planes. This was met with varying degrees of success but largely the method was sound with some occasional algebraic slips. From the Cartesian equation, most could identify the position and direction of the line correctly. A significant number of students correctly used a vector product to find the direction of the line.

In question (b) students with a vector equation from question (a) could proceed correctly and substituted into the third plane to identify the intersection of the 3 planes. Some students chose to solve 3 simultaneous equations to find the required point and were, in many cases, successful.

Question Number	Scheme	Marks
(a)	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 3 \\ -1 & 2 & 4 \end{vmatrix} = \begin{pmatrix} -2 \\ -11 \\ 5 \end{pmatrix}$	M1A1
	M1: Attempt cross product of normal vectors. If method unclear, 2 components must be correct.	
	A1: Correct vector	
	$x = 0: (0, \frac{1}{2}, \frac{3}{2}), y = 0: (-\frac{1}{11}, 0, \frac{10}{11}), z = 0: (\frac{2}{5}, \frac{19}{5}, 0)$	M1A1
	M1: Attempt point on the line (x, y and z). A1: Correct coordinates	
	$\mathbf{r} = \frac{1}{2}\mathbf{j} + \frac{3}{2}\mathbf{k} + \lambda(2\mathbf{i} + 11\mathbf{j} - 5\mathbf{k})$	ddM1A1
	M1: Their point + λ their direction Dependent on both previous method marks.	
	A: Correct equation (oe)	
	(6)	
	Alternative 1	
	$x = \frac{y - \frac{1}{2}}{\frac{11}{2}} = \frac{z - \frac{3}{2}}{-\frac{5}{2}} \text{ or } \frac{x + \frac{1}{11}}{\frac{2}{11}} = y = \frac{z - \frac{10}{11}}{-\frac{5}{11}} \text{ or } \frac{x - \frac{3}{5}}{-\frac{2}{5}} = \frac{y - \frac{19}{5}}{-\frac{11}{5}} = z$	M1A1
	M1: Correctly attempts cartesian equations of line A1: Correct equations	
	$(0, \frac{1}{2}, \frac{3}{2}) \text{ or } (-\frac{1}{11}, 0, \frac{10}{11}) \text{ or } (\frac{2}{5}, \frac{19}{5}, 0)$	
	M1: Extracts position correctly A1: Correct position or $\lambda(2\mathbf{i} + 11\mathbf{j} - 5\mathbf{k})$	M1A1
	M1: Extracts direction correctly A1: Correct direction	
	$\mathbf{r} = \frac{1}{2}\mathbf{j} + \frac{3}{2}\mathbf{k} + \lambda(2\mathbf{i} + 11\mathbf{j} - 5\mathbf{k})$	ddM1A1
	M1: Their point + λ their direction Dependent on both previous method marks.	
	A: Correct equation (oe)	
	Alternative 2	
	$x = \lambda \Rightarrow y = \frac{11\lambda + 1}{2}, z = \frac{3 - 5\lambda}{2} \text{ (oe)}$	M1A1
	M1: Obtains x, y and z in terms of " λ " A1: Correct expressions	
	$(0, \frac{1}{2}, \frac{3}{2})$ or $\lambda(2\mathbf{i} + 11\mathbf{j} - 5\mathbf{k})$	M1A1
	M1: Extracts position correctly A1: Correct position or M1: Extracts direction correctly A1: Correct direction	
	$\mathbf{r} = \frac{1}{2}\mathbf{j} + \frac{3}{2}\mathbf{k} + \lambda(2\mathbf{i} + 11\mathbf{j} - 5\mathbf{k})$	ddM1A1
	M1: Their point + λ their direction Dependent on both previous method marks.	
	A: Correct equation (oe)	
(b)	$2\lambda - (\frac{1}{2} + 11\lambda) + 2(\frac{3}{2} - 5\lambda) = 31$	M1
	$\lambda = \frac{-3}{2}$	
	Planes intersect at $(-3, -16, 9)$	M1A1
	M1: Substitutes into their line A1: Correct coordinates	
	(3)	
	Alternative	
	M1: Solves three simultaneous equations to obtain one value for x or y or z M1: Solves three simultaneous equations to obtain values for x, y and z A1: Correct coordinates	
	Total 9	

Q5. Part of the mains water system for a housing estate consists of water pipes buried beneath the ground surface. The water pipes are modelled as straight line segments. One water pipe, W , is buried beneath a

particular road. With respect to a fixed origin O , the road surface is modelled as a plane with equation $3x - 5y - 18z = 7$, and W passes through the points $A(-1, -1, -3)$ and $B(1, 2, -3)$. The units are in metres.

(a) Use the model to calculate the acute angle between W and the road surface.

(5)

A point $C(-1, -2, 0)$ lies on the road. A section of water pipe needs to be connected to W from C .

(b) Using the model, find, to the nearest cm, the shortest length of pipe needed to connect C to W .

(6)

(a) Most students appreciated the strategy needed to solve the problem and use the scalar product to find an angle between the normal vector and the direction of W . Unfortunately, many could not then find the required acute angle – a clear diagram would have helped them to visualise which angle was required. A common error was to subtract the result of their inverse cosine from 180° . Those using sine were usually more successful here. A few found the correct angle but failed to give it to the correct accuracy or to indicate degrees or radians and hence lost the final mark.

(b) The majority of students formed the correct parametric form for W but many did not then find CW and some were attempting a scalar product with a direction vector perpendicular to the plane. There were often errors with signs for those who were working with the appropriate vectors. Those with the correct CW were usually then able to accurately use Pythagoras' Theorem to gain full marks here but they were in a minority.

Some of the students who correctly found the value of their parameter, substituted this into the equation of line W and found the magnitude of this vector rather than finding the vector CW and its magnitude.

Those students who produced a diagram of the situation were more successful. It is perhaps a pity that a number having done fully correct work lost the final mark as they either failed to include units or thought that the units were centimetres throughout.

Question	Scheme	Marks	AOs
(a)	Attempts the scalar product between the direction of W and the normal to the road and uses trigonometry to find an angle.	M1	3.1a
	$\left(\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 \\ -5 \\ -18 \end{pmatrix} = -9$ or $\left(\begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 \\ -5 \\ -18 \end{pmatrix} = 9$	M1 A1	1.1b 1.1b
	$\sqrt{(2)^2 + (3)^2 + (0)^2} \sqrt{(3)^2 + (-5)^2 + (-18)^2} \cos \alpha = "-9"$ $\theta = 90 - \arccos\left(\frac{9}{\sqrt{13}\sqrt{358}}\right)$ or $\theta = \arcsin\left(\frac{9}{\sqrt{13}\sqrt{358}}\right)$ Angle between pipe and road = 7.58° (3sf) or 0.132 radians (3sf) (Allow -7.58° or -0.132 radians)	M1 A1	1.1b 3.2a
		(5)	

(b)	$W: \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix} + t \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$	B1ft	1.1b
	$C \text{ to } W: \left\{ \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix} + t \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix} \right\} \text{ or } \left\{ \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix} \right\}$	M1	3.4
	$\begin{pmatrix} 2t \\ 3t+1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = 0 \Rightarrow t = \dots \text{ or } \begin{pmatrix} 2+2\lambda \\ 4+3\lambda \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = 0 \Rightarrow \lambda = \dots$ or $(2t)^2 + (3t+1)^2 + (-3)^2 = \dots \text{ or } (2+2\lambda)^2 + (4+3\lambda)^2 + (-3)^2 = \dots$	M1	3.1b
	$t = -\frac{3}{13} \text{ or } \lambda = -\frac{16}{13} \Rightarrow (C \text{ to } W)_{\min} \text{ is } -\frac{6}{13}\mathbf{i} + \frac{4}{13}\mathbf{j} - 3\mathbf{k}$ or $(2t)^2 + (3t+1)^2 + (-3)^2 = 13\left(t + \frac{3}{13}\right)^2 + \frac{121}{13}$ or $(2+2\lambda)^2 + (4+3\lambda)^2 + (-3)^2 = 13\left(\lambda + \frac{16}{13}\right)^2 + \frac{121}{13}$ or $\frac{d((2t)^2 + (3t+1)^2 + (-3)^2)}{dt} = 0 \Rightarrow t = -\frac{3}{13} \Rightarrow C \text{ to } W \text{ is } -\frac{6}{13}\mathbf{i} + \frac{4}{13}\mathbf{j} - 3\mathbf{k}$ Or $\frac{d((2+2\lambda)^2 + (4+3\lambda)^2 + (-3)^2)}{d\lambda} = 0 \Rightarrow \lambda = -\frac{16}{13} \Rightarrow (C \text{ to } W)_{\min} \text{ is } -\frac{6}{13}\mathbf{i} + \frac{4}{13}\mathbf{j} - 3\mathbf{k}$	A1	1.1b
	$d = \sqrt{\left(-\frac{6}{13}\right)^2 + \left(\frac{4}{13}\right)^2 + (-3)^2} \text{ or } d = \sqrt{\frac{121}{13}}$	ddM1	1.1b

Shortest length of pipe needed is 305 or 305 cm or 3.05 m	A1	3.2a
	(6)	

(11 marks)

Notes

(a)

M1: Realises the scalar product between the direction of W and the normal to the road is needed and so applies it and uses trigonometry to find an angle

M1: Calculates the scalar product between $\pm \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix}$ and $\pm \begin{pmatrix} 3 \\ -5 \\ -18 \end{pmatrix}$ (Allow sign slips as

long as the intention is clear)

A1: $\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -18 \end{pmatrix} = -9$ or $\begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -18 \end{pmatrix} = 9$ or $\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 5 \\ 18 \end{pmatrix} = 9$ or $\begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 5 \\ 18 \end{pmatrix} = -9$

M1: A fully complete and correct method for obtaining the acute angle

A1: Awrt 7.58° or awrt 0.132 radians (must see units). Do not isw and withhold this mark if extra answers are given.

(b)

B1ft: Forms the correct parametric form for the pipe W . Follow through their direction vector for W from part (a).

M1: Identifies the need to and forms a vector connecting C to W using a parametric form for W

M1: Uses the model to form the scalar product of C to W and the direction of W to find the value of their parameter or finds the distance C to W or $(C \text{ to } W)^2$ in terms of their parameter

A1: Correct vector or correct completion of the square

ddM1: Correct use of Pythagoras on their vector CW or appropriate method to find the shortest distance between the point and the pipe. Dependent on both previous method marks.

A1: Correct length for the required section of pipe is 305 or 305 cm or 3.05 m

Alternatives for part (b):

(b) Way 2	$AC = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}, AB = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$	B1ft	1.1b
	$AC \cdot AB = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = 3$	M1	3.4
	$\Rightarrow \cos CAB = \frac{3}{\sqrt{10}\sqrt{13}} \Rightarrow CAB = \dots$	M1	3.1b
	$CAB = 74.74\dots^\circ$	A1	1.1b
	$d = \sqrt{10} \sin 74.74\dots^\circ$	ddM1	1.1b
	Shortest length of pipe needed is 305 or 305 cm or 3.05 m	A1	3.2a
		(6)	

Notes			
(b) B1ft: Forms the correct vectors. Follow through their direction vector for W from part (a). M1: Identifies the need to and forms the scalar product between AC and AB M1: Uses the model to form the scalar product and uses this to find the angle CAB A1: Correct angle ddM1: Correct method using their values or appropriate method to find the shortest distance between the point and the pipe. Dependent on both previous method marks. A1: Correct length for the required section of pipe is 305 or 305 cm or 3.05 m			

(b) Way 3	$AC = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}, AB = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$	B1ft	1.1b
	$AC \times AB = \begin{vmatrix} i & j & k \\ 0 & -1 & 3 \\ 2 & 3 & 0 \end{vmatrix} = \begin{pmatrix} -9 \\ 6 \\ 2 \end{pmatrix}$	M1	3.4
	$ AC \times AB = \sqrt{9^2 + 6^2 + 2^2} = \dots$	M1	3.1b
	$= 11$	A1	1.1b
	$d = \frac{11}{ AB } = \frac{11}{\sqrt{2^2 + 3^2}} = \dots$	ddM1	1.1b
	Shortest length of pipe needed is 305 or 305 cm or 3.05 m	A1	3.2a
		(6)	

Notes			
(b) B1ft: Forms the correct vectors. Follow through their direction vector for W from part (a). M1: Identifies the need to and forms the vector product between AC and AB M1: Uses the model to find the magnitude of their vector product A1: Correct value ddM1: Correct method using their values or appropriate method to find the shortest distance between the point and the pipe. Dependent on both previous method marks. A1: Correct length for the required section of pipe is 305 or 305 cm or 3.05 m			